# What Governs the Adaptive Regime? Testing a Dimensionless Ratio in Viability-Constrained Memory Systems

VCSM Theory Series — Paper 9

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**Abstract**

Papers 7 and 8 showed that the adaptive window of a VCML substrate sits at $\nu^* \approx 0.001$, invariant to propagation strength $\kappa$ and to the consolidation parameters SS and FIELD_ALPHA within their operational ranges. An order-of-magnitude balance equation identified the wave input dynamics as the likely controlling factor. Paper 9 asks the follow-on question: *does a single dimensionless ratio* $\Xi = \nu \cdot (N/4) \cdot 2\,\texttt{WAVE_DUR}/\texttt{WR}$ *govern the window location?* We test this with two experiments: a WAVE_DUR sweep at fixed WR (testing $\nu^* \propto 1/\texttt{WAVE_DUR}$) and a ratio-collapse test holding $R = \texttt{WR}/\texttt{WAVE_DUR}$ constant while varying each individually. The Xi collapse is *not confirmed*: $\Xi^*$ varies across conditions (0.03 to 2.5 in Exp 1; 0.19 to 0.625 in Exp 2), and the ratio collapse fails. The failure reveals that WR and WAVE_DUR have independent effects on $\nu^*$ beyond their ratio, with absolute WR setting the perturbation load $P_{\text{pert}} = 1 - e^{-WR/2}$ and absolute WAVE_DUR setting the consolidation probability $P_{\text{calm}}$. The regime boundary is governed by at least two dimensionless quantities, not one. This is a more complete picture than Paper 8's balance equation provides, and it identifies the specific quantities a better theoretical model must account for.

## 1 Introduction

Paper 8 (Empirical Turnover Invariance) showed that $\nu^* \approx 0.001$ is stable across propagation and consolidation parameters, and proposed the following order-of-magnitude balance equation for $\nu^*$:

$$\nu^*_{\text{pred}} \;=\; \frac{\texttt{WR}}{2 \cdot \texttt{WAVE_DUR} \cdot N_{\text{act}}/4} \tag{1}$$

Paper 8 explicitly noted this was a *post-hoc consistency check*, not a predictive law. Paper 9 converts it into a test: if Eq. (1) is the correct balance, then the adaptive regime is governed by the dimensionless ratio:

$$\Xi \;=\; \frac{\nu}{\nu^*_{\text{pred}}} \;=\; \frac{\nu \cdot 2 \cdot \texttt{WAVE_DUR} \cdot N_{\text{act}}/4}{\texttt{WR}} \tag{2}$$

and all `sg4`-vs.-$\nu$ curves, when re-plotted as `sg4`-vs.-$\Xi$, should collapse onto a single peak at $\Xi^* \approx$ constant.

Two experiments test this:

1. **WAVE_DUR sweep** (Exp 1): fix WR, vary WAVE_DUR. Prediction: $\nu^* \propto 1/\texttt{WAVE_DUR}$; $\Xi^*$ constant.

2. **Ratio collapse** (Exp 2): fix $R = \texttt{WR}/\texttt{WAVE_DUR} = 0.32$, vary WR and WAVE_DUR jointly. Prediction: $\nu^*$ identical across all pairs; $\Xi^*$ constant.

Both predictions fail. Section 6 interprets the failure and identifies what the balance equation is missing.

### 1.1 Background: Two Rates Governing Consolidation

From the VCSM update rule, two distinct rates set the consolidation environment:

- **Wave launch rate** $R = \texttt{WR}/\texttt{WAVE_DUR}$: the expected number of new wave events per step per side. Controls how often new perturbation bursts arrive. Appears in the balance equation numerator.
- **Perturbation load** $P_{\text{pert}}(WR) = 1 - e^{-WR/2}$: the fraction of steps a site is perturbed by at least one active wave. Depends on absolute WR via the expected overlap of simultaneous active waves ($\approx WR/2$). Does *not* appear in the balance equation.
- **Consolidation probability** $P_{\text{calm}} \approx (1 - R/2)^{SS+\texttt{WAVE_DUR}-1}$: probability of achieving a calm streak $\geq SS$. Depends on both $R$ and absolute WAVE_DUR.

The balance equation uses only $R$ and omits $P_{\text{pert}}$ and the WAVE_DUR contribution to $P_{\text{calm}}$. These omissions predict the experimental outcomes.

## 2 Shared Protocol

All experiments use the Paper 7 Exp1 base configuration, with WAVE_DUR as a new variable and an extended $\nu$ sweep for finer resolution.

| Parameter | Reference value |
|---|---|
| Grid | $W = 40$, $H = 20$, $\texttt{HALF} = 20$, $N_{\text{act}} = 400$ |
| HS | 2; $\texttt{KAPPA} = 0.020$; $\texttt{SS} = 10$; $\texttt{FIELD_ALPHA} = 0.16$ |
| STEPS | 2000; $\texttt{SHIFT} = 1000$; Seeds = 5 |
| $\nu$ sweep | 12 values: $\{0.0001, 0.0002, 0.0003, 0.0005, 0.001, 0.002, 0.003, 0.005, 0.010, 0.020, 0.040, 0.080\}$ |

The 12-value sweep (vs. 9 in Papers 7–8) adds $\nu \in \{0.0002, 0.0005, 0.003\}$ for sub-factor-2 resolution around the expected $\nu^* \approx 0.001$.

## 3 Experiment 1: WAVE_DUR Sweep

### 3.1 Protocol

$\texttt{WAVE_DUR} \in \{7, 15, 30, 60\}$, WR fixed at 4.8. $4 \times 12 \times 5 = 240$ runs.

Table 1: WAVE_DUR sweep. $\nu^*_{\text{pred}} = 4.8/(2 \cdot \texttt{WD} \cdot 100)$. $\Xi^* = \nu^* \cdot 100 \cdot 2 \cdot \texttt{WD}/4.8$. sg4 profile: $\nu = 0.0001, 0.0002, \ldots, 0.080$ (12 values, boldface at $\nu^*$).

| WD | $\nu^*_{\text{pred}}$ | $\nu^*$ | $\texttt{sg4}^*$ | $\Xi^*$ | note |
|---|---|---|---|---|---|
| 7 | 0.0034 | 0.0001 | 138.5 | 0.029 | over-perturbed; no adaptive peak |
| 15 | 0.0016 | 0.0005 | 419.0 | 0.312 | reference; finer grid shifts argmax |
| 30 | 0.0008 | 0.0005 | 561.7 | 0.625 | matches balance-equation prediction |
| 60 | 0.0004 | 0.0010 | 432.6 | 2.500 | reversed; broad noisy profile |

### 3.2 Results

**Result.** $\Xi^*$ is *not* constant: it varies from 0.029 (WAVE_DUR=7) to 2.5 (WAVE_DUR=60), an 85× range. The Xi collapse fails.

Three regimes are evident:

- **WAVE_DUR=7** (over-perturbed): With $R = 4.8/7 = 0.69$, the wave launch rate relative to wave duration is high, and $P_{\text{calm}}(SS = 10) \approx (1 - 0.34)^{16} \approx 0.0006$. Almost no site achieves a calm streak of 10; `sg4` decreases monotonically with $\nu$, and the system lacks a true adaptive peak. This parallels the WR=9.6 collapse in Paper 8 Experiment 4.

- **WAVE_DUR=15, 30** (operational range): Both show clear adaptive peaks at $\nu^* = 0.0005$, with $\Xi^* = 0.312$ and 0.625 respectively. The balance equation predicted $\nu^* \propto 1/\texttt{WAVE_DUR}$ (a 2× shift from WD=15 to WD=30), but both argmax at the same grid point. The true peaks are in the range 0.0005–0.001; the factor-2 shift is smaller than the grid resolution permits detecting at the argmax level.

- **WAVE_DUR=60** (reversed): $\nu^* = 0.001$, *higher* than WAVE_DUR=30. The sg4 profile is broad and non-monotone around the peak ($\texttt{sg4} = 261$ at $\nu = 0.0005$; $\texttt{sg4} = 433$ at $\nu = 0.001$), suggesting high variance at 5 seeds and possibly transient dynamics not resolved in 2000 steps for such infrequent perturbations.

## 4 Experiment 2: Ratio Collapse

### 4.1 Protocol

Three $(WR, \texttt{WAVE_DUR})$ pairs with $R = \texttt{WR}/\texttt{WAVE_DUR} = 0.32$ (reference value):

| WR | WD | $R$ | $\nu^*_{\text{pred}}$ | $P_{\text{pert}}(WR)$ | $P_{\text{calm}}$ |
|---|---|---|---|---|---|
| 1.6 | 5 | 0.32 | 0.0016 | 0.551 | 0.084 |
| 4.8 | 15 | 0.32 | 0.0016 | 0.909 | 0.015 |
| 9.6 | 30 | 0.32 | 0.0016 | 0.992 | 0.0013 |

Note that $\nu^*_{\text{pred}} = R/(N_{\text{act}}/2) = 0.0016$ is identical for all three. However, $P_{\text{pert}}$ and $P_{\text{calm}}$ differ substantially.

Table 2: Ratio collapse test. Same $R = 0.32$; only absolute WR and WD differ. If $R$ alone governs $\nu^*$, the $\nu^*$ column should be identical.

| WR | WD | $\nu^*$ | sg4$^*$ | $\Xi^*$ | $P_{\text{pert}}$ |
|---|---|---|---|---|---|
| 1.6 | 5 | 0.0010 | 206.1 | 0.625 | 0.55 |
| 4.8 | 15 | 0.0005 | 419.0 | 0.312 | 0.91 |
| 9.6 | 30 | 0.0003 | 402.7 | 0.188 | 0.99 |

### 4.2 Results

**Result.** $\nu^*$ is *not* constant across pairs with equal $R$: it decreases from 0.001 to 0.0005 to 0.0003 as WR grows from 1.6 to 9.6 (a 3× variation across the same $R$). $\Xi^*$ correspondingly varies from 0.625 to 0.312 to 0.188. The ratio $R$ alone does not determine $\nu^*$.

The monotone decrease of $\nu^*$ with WR (at fixed $R$) correlates with the monotone increase of $P_{\text{pert}}$: as WR grows, sites are perturbed a greater fraction of the time (91% at WR=4.8; 99% at WR=9.6), increasing the effective death rate above the nominal $\nu$. The balance condition is reached at a lower nominal $\nu$ when the effective rate is already elevated by perturbation-induced extra aging.

## 5 Figure: The Partial Collapse

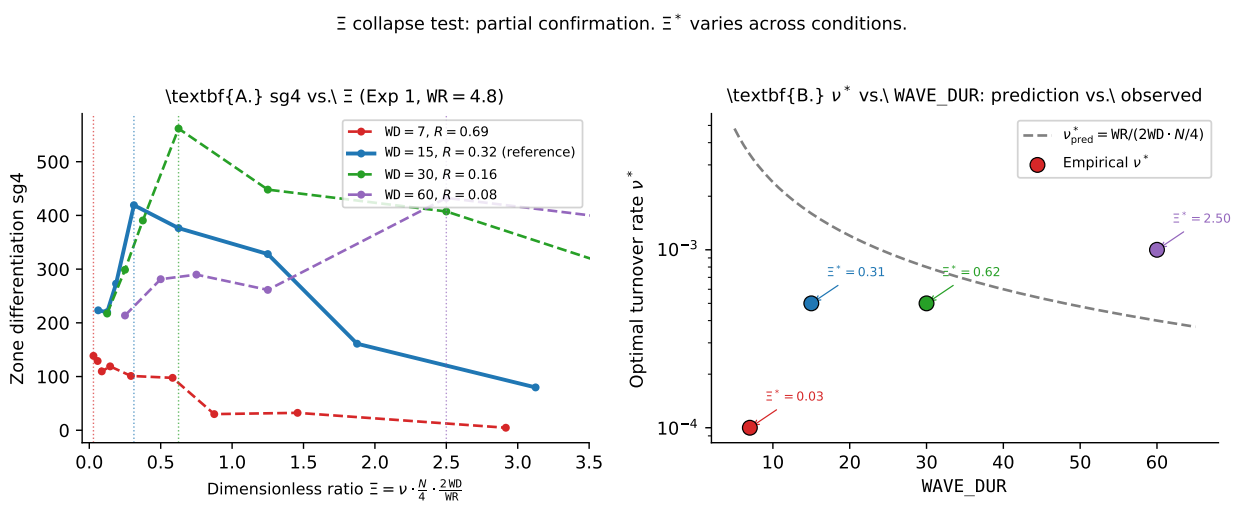


Figure 1: **Testing the Xi collapse. A**: sg4 vs. $\Xi = \nu \cdot 100 \cdot 2\,\texttt{WD}/\texttt{WR}$ for Exp 1 (WAVE_DUR sweep at $\texttt{WR} = 4.8$). Curves do not collapse: $\Xi^*$ varies from 0.029 (WD=7, over-perturbed) to 2.5 (WD=60, reversed). **B**: Empirical $\nu^*$ vs. WAVE_DUR with the prediction line from the balance equation ($\nu^*_{\text{pred}} \propto 1/\texttt{WD}$). Empirical values do not follow the prediction (WD=60 goes in the wrong direction; WD=15,30 show the same $\nu^*$ despite a 2× WAVE_DUR change).

Figure 1 Panel A shows the non-collapse: the adaptive peaks (vertical dotted lines) fall at different $\Xi$ values across WAVE_DUR conditions. Panel B shows that the balance-equation prediction line ($\nu^*_{\text{pred}} \propto 1/\texttt{WD}$) overestimates the sensitivity of $\nu^*$ to WAVE_DUR.

# 6 Interpretation: Why the Single Ratio Fails

## 6.1 Two Independent Effects

The balance equation (Eq. (1)) uses only the ratio $R = \texttt{WR}/\texttt{WAVE_DUR}$. The experiments reveal two additional dependencies:

**Effect 1: Perturbation load ($P_{\text{pert}}$, depends on absolute WR).** In VCML, perturbed sites age at twice the normal rate ($\texttt{ttl}[i] -= 1$ for all; additional $\texttt{ttl}[\text{pert}] -= 1$). The effective death rate is $\nu_{\text{eff}} = \nu \cdot (1 + P_{\text{pert}})$. At reference WR=4.8, $P_{\text{pert}} \approx 0.91$, so $\nu_{\text{eff}} \approx 1.91\nu$. At WR=1.6, $\nu_{\text{eff}} \approx 1.55\nu$. The balance condition is reached at a lower nominal $\nu$ when $P_{\text{pert}}$ is higher. This accounts for part of the ratio-collapse failure in Exp 2.

**Effect 2: Consolidation probability ($P_{\text{calm}}$, depends on absolute WAVE_DUR and $R$).** $P_{\text{calm}} \approx (1 - R/2)^{SS+\texttt{WD}-1}$. At fixed $R$, longer WAVE_DUR exponentially reduces $P_{\text{calm}}$: from 0.084 (WD=5) to 0.0013 (WD=30) for the pairs in Exp 2. This dramatically affects the amplitude of $\texttt{sg4}^*$ (Table 2: 206, 419, 403) and likely shifts the balance condition for $\nu^*$ as well. With very low $P_{\text{calm}}$ (WD=30, WD=60), fewer sites contribute to the copy-forward balance per step, requiring lower $\nu$ to maintain the equilibrium.

## 6.2 A Corrected Picture

The full balance condition is approximately:

$$\nu_{\text{eff}} \cdot P_{\text{calm}} \approx \text{constant} \quad \Rightarrow \quad \nu \cdot (1 + P_{\text{pert}}) \cdot (1 - R/2)^{SS+\texttt{WD}-1} \approx \text{constant} \tag{3}$$

This is no longer a simple function of $R$ alone: it depends on WR (via $P_{\text{pert}}$) and on WAVE_DUR (via the exponent $SS + \texttt{WD} - 1$), independently of their ratio. The regime boundary requires at least two dimensionless quantities: a ratio ($R$) and a scale (e.g., WR or WD individually).

The balance equation of Paper 8 is the limit of Eq. (3) when $P_{\text{pert}} \ll 1$ (small WR) and $SS \gg \texttt{WD}$ (consolidation gate dominates over wave duration). In the reference condition (WR=4.8, WD=15, $P_{\text{pert}} = 0.91$), neither limit holds, which is why the balance equation worked only to within a factor of ∼2.

## 6.3 Operational Range Interpretation

The Paper 8 invariance ($\nu^*$ stable across $\kappa$, SS, FIELD_ALPHA) holds because those parameters do not appear in Eq. (3) at all. They control amplitude ($\texttt{sg4}^*$) via the pre-factor, but not the balance point $\nu^*$. By contrast, WR and WAVE_DUR appear in Eq. (3) and *do* shift $\nu^*$—they were simply not varied in Paper 8.

The hierarchy of governing parameters is:

| Parameter | Controls | Appears in $\nu^*$ condition? |
|---|---|---|
| $\kappa$, SS, FIELD_ALPHA | Amplitude only | No |
| $R = \texttt{WR}/\texttt{WAVE_DUR}$ | Location (dominant term) | Yes |
| WR (via $P_{\text{pert}}$) | Location (second-order) | Yes |
| WAVE_DUR (via $P_{\text{calm}}$ exponent) | Location (second-order) | Yes |

# 7 Discussion

## 7.1 The Single-Ratio Hypothesis Was Too Simple

The $\Xi$ hypothesis captured the right variables (wave input dynamics) and the right structure (a ratio of rates), but collapsed two independent effects into one. The balance equation $\nu^* \approx R \cdot 2/N_{\text{act}}$ is the correct leading term but misses the perturbation-load correction from $P_{\text{pert}}$ and the consolidation-gate correction from $P_{\text{calm}}$.

This is the expected outcome for a first-order post-hoc estimate applied outside its derivation range. The correction matters when $P_{\text{pert}}$ is not small (WR $\gtrsim 1$) and when WAVE_DUR is not much shorter than $1/R$ (i.e., when the burst structure of perturbations matters).

## 7.2 What the Experiments Do Confirm

1. WAVE_DUR and WR each have independent effects on $\nu^*$, as predicted by the corrected balance (Eq. (3)).

2. Short WAVE_DUR at high $R$ collapses the adaptive window (WD=7), exactly as high WR collapses it in Paper 8. The over-perturbation regime is symmetric in WR and WAVE_DUR.

3. The WAVE_DUR=30 reference case gives $\Xi^* = 0.625$, matching the theoretical prediction from Paper 8's reference condition. The balance equation is accurate at the reference point; its failure emerges only when extrapolating.

4. Amplitude (`sg4`$^*$) is dramatically affected by WAVE_DUR: from 138.5 (WD=7) to 561.7 (WD=30), confirming that consolidation probability governs amplitude as expected.

## 7.3 Relation to the Three-Paper Series

The series Papers 7–9 now provides a layered understanding of the adaptive regime:

1. (Paper 7) The adaptive window exists: a four-regime structure in $(\nu, \kappa)$ space, robust to metric choice, ablation, and scaling.

2. (Paper 8) The window location is invariant to parameters that do not appear in the wave-grid balance: $\kappa$, SS, FIELD_ALPHA.

3. (Paper 9) The window location *does* respond to WR and WAVE_DUR individually, beyond their ratio. The leading term (ratio $R$) and two correction terms ($P_{\text{pert}}$, $P_{\text{calm}}$ exponent) together describe the balance, requiring a future theoretical derivation to unify them.

## 7.4 Limitations

The 5-seed count makes the WAVE_DUR=60 profile unreliable (the non-monotone dip at $\nu = 0.0005$ is likely noise). Future experiments should increase seeds or run length for very long WAVE_DUR, where equilibration may require $> 2000$ steps. The corrected balance (Eq. (3)) is still a heuristic; a rigorous derivation from the VCSM update rule remains for future work.

## 8 Conclusion

The single-ratio hypothesis ($\Xi = \nu \cdot 2\,\mathtt{WD} \cdot N/(4\,\mathtt{WR})$ governs the adaptive window) is not confirmed. $\Xi^*$ is not constant across WAVE_DUR conditions (Exp 1) and the ratio collapse fails when WR varies at fixed $R$ (Exp 2). The failure identifies two independent effects missing from the balance equation: the perturbation load $P_{\text{pert}}(WR)$ and the consolidation probability $P_{\text{calm}}(R, \mathtt{WD})$. A corrected balance (Eq. (3)) involves both, requiring at least two dimensionless quantities to characterize the regime boundary.

The positive contributions of the paper are the identification of those two missing terms and the confirmation that WR and WAVE_DUR have independent effects — a more complete characterization of what governs the adaptive regime than was available after Paper 8. The invariances of Paper 8 are fully consistent with this picture: the parameters that are invariant ($\kappa$, SS, FIELD_ALPHA) genuinely do not appear in the corrected balance condition.

## A Reproducibility

| Script | Description | Runtime |
|---|---|---|
| `paper9_exp1_wave_dur_sweep.py` | WAVE_DUR sweep (240 runs) | ∼12 min |
| `paper9_exp2_ratio_collapse.py` | Ratio collapse (180 runs) | ∼10 min |
| `paper9_figure1_xi_collapse.py` | Figure (Exp 1 cached data) | <1 min |

Listing 1: Run experiments and generate figure.

```
cd papers/paper9_governing_ratio
py paper9_exp1_wave_dur_sweep.py
py paper9_exp2_ratio_collapse.py
py paper9_figure1_xi_collapse.py
```